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A

Mini Project Report

On

AGRICONNECT- DIRECT TRADE

Submitted By

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ABSTRACT

AgriConnect is a cloud-native digital marketplace designed to connect farmers directly with buyers through a transparent and secure trading platform. The system eliminates intermediaries, enables real-time bidding and fixed-price listings, and ensures secure transactions using escrow-based payment mechanisms.

Built using React (Next.js), Supabase, PostgreSQL, and WebSockets, the platform supports real-time communication, automated price discovery, and scalable cloud deployment.

AgriConnect increases farmer income, reduces food waste, and supports SDG 2 (Zero Hunger) and SDG 12 (Responsible Consumption and Production) by optimizing agricultural trade and improving supply chain efficiency.

PROBLEM STATEMENT

In the current agricultural marketing system, farmers heavily depend on middlemen to sell their produce. Due to a lack of direct market access, price transparency, and digital awareness, farmers often receive lower profits for their crops. Buyers, on the other hand, may pay higher prices due to intermediary commissions.

Additionally, farmers lack real-time information about market prices and weather conditions, which affects productivity and income stability. There is no centralized digital platform that enables direct interaction between farmers and buyers in a transparent and secure manner.

Solution – AGRI CONNECT DIRECT TRADE

To overcome the challenges faced by farmers in the traditional agricultural system, Agri Connect proposes a digital marketplace platform that connects farmers directly with buyers.

- **Direct Farmer-to-Buyer Marketplace**

  The platform allows farmers to register and list their crops with details such as price, quantity, and location. Buyers can search and purchase crops directly without middlemen involvement. This ensures fair pricing and better profit margins for farmers.

- **Transparent Pricing System**

  Farmers can set their own prices based on market trends. Buyers can compare prices and choose the best option, ensuring transparency and competitive trade.

- **Real-Time Information Access**

  The system integrates weather updates and market information to help farmers make informed decisions regarding harvesting and selling.

- **Secure and Digital Transactions**

  The platform includes secure login authentication, encrypted passwords, and order tracking to ensure safe and reliable transactions.

- **Scalable and Future-Ready Architecture**

  The system is built using a three-tier architecture (Frontend, Backend, Database), making it scalable and suitable for future enhancements like AI-based price prediction, online payment gateways, and mobile app integration.

Technologies Used

Frontend:

React.js
Next.js Framework
Tailwind CSS

Backend:

Supabase (Backend-as-a-Service)
Supabase Authentication

Database:

PostgreSQL (Cloud Database via Supabase)
Real-Time Communication
WebSockets (for live bidding and instant updates)

Outcome

- **Increased Farmer Income**
  By eliminating middlemen, farmers can sell their crops directly to buyers and receive better prices, improving their overall profit margins.
- **Transparent Pricing System**
  Buyers can view crop prices directly from farmers, ensuring fair trade practices and price transparency in the marketplace.
- **Improved Market Access**
  Farmers gain access to a wider customer base beyond local markets, expanding their business opportunities.
- **Time-Saving and Efficient Process**
  Digital crop listing and online order placement reduce the time spent in traditional market transactions.
- **Real-Time Information Availability**
  Weather updates and market data help farmers make informed decisions regarding harvesting and selling.
- **Secure and Organized Data Management**
  The system securely stores user and transaction data, improving record maintenance and reducing manual errors.
- **Digital Empowerment of Rural Communities**
  The platform encourages technology adoption in rural areas and promotes digital literacy among farmers.
- **Foundation for Future Enhancements**
  The system architecture supports future upgrades such as AI- based crop prediction, online payment gateways, and mobile application integration.

CHAPTER 1: INTRODUCTION

Background

Agriculture is one of the most important sectors of India's economy, contributing significantly to employment and Gross Domestic Product (GDP). A large percentage of the population depends on farming as their primary source of livelihood. However, despite their contribution, many farmers face financial challenges due to inefficient marketing systems, lack of price transparency, and heavy dependence on intermediaries.

Traditionally, farmers sell their produce in local markets where middlemen often control pricing and distribution. Due to limited access to real-time market information, digital tools, and broader buyer networks, farmers frequently sell their crops at lower prices. With the rapid growth of digital technology and internet connectivity, there is a strong need for a centralized, cloud-based online platform that connects farmers directly with buyers.

AgriConnect is developed as a modern digital marketplace to transform agricultural trade by ensuring transparency, efficiency, secure transactions, and fair price discovery.

Problem Statement

In the existing agricultural marketing system, farmers rely heavily on intermediaries to sell their produce. This dependency results in reduced profit margins and limited bargaining power. The absence of transparent pricing mechanisms and real-time market data prevents farmers from making informed selling decisions.

The key problems include:

- Low profit margins for farmers due to intermediary commissions
- Lack of transparent and real-time price discovery
- Limited direct communication between farmers and buyers
- Absence of a centralized digital marketplace
- Inefficient supply chain coordination
- Limited access to real-time demand and market insights

Buyers also face challenges such as higher procurement costs, inconsistent pricing, and difficulty in sourcing produce directly from verified farmers.

Objectives

The primary objectives of AgriConnect are:

- To develop a cloud-based digital marketplace connecting farmers and buyers
- To eliminate dependency on middlemen in agricultural trade
- To increase farmers' income through direct selling and real-time bidding
- To enable transparent price discovery mechanisms
- To ensure secure authentication and transaction processes
- To provide a scalable and reliable cloud infrastructure
- To promote digital empowerment in rural communities

Scope of the Project:

The scope of AgriConnect includes:

- Farmer registration and secure authentication
- Product (crop) listing with pricing options (fixed price or auction)
- Buyer registration and product browsing
- Real-time bidding system using WebSockets
- Order placement and tracking
- Secure payment workflow with transaction records
- Cloud-based database management (PostgreSQL)
- Admin monitoring and system management
- Real-time data synchronization across users

Future Scope

The future enhancements of AgriConnect may include:

- Full escrow-based integrated payment gateway
- AI-based crop price prediction using demand analytics
- Predictive demand forecasting models
- Mobile application development (Android/iOS)
- Multi-language support for regional accessibility
- Blockchain-based settlement layer for enhanced transparency
- Advanced analytics dashboard for farmers and buyers

CHAPTER 2: LITERATURE SURVEY

1.Traditional Agricultural Marketing Systems

Traditional agricultural marketing systems in India are primarily based on physical marketplaces known as mandis, where farmers sell their produce through intermediaries such as commission agents and wholesalers. In this system, middlemen play a dominant role in price negotiation and distribution. Several studies indicate that intermediaries capture a significant share of the final selling price, resulting in reduced income for farmers.

These conventional systems lack transparency, digital record management, and real-time access to market information. Farmers often depend on local demand and have limited bargaining power due to insufficient price awareness. Additionally, the absence of integrated weather forecasting and demand analysis tools contributes to inefficiencies and post-harvest losses.

2.Existing Reporting Applications for Agriculture

Various applications and tools have been developed to support farmers, mostly focusing on isolated functionalities such as weather updates or price information:

a. Market Price Reporting Systems

Platforms such as Agmarknet and other mandi price portals provide periodic updates on crop prices across regions. While these systems improve awareness of prevailing market rates, they function primarily as informational portals. They do not enable direct transactions between farmers and buyers. Moreover, price updates may not always be real-time, and there is no built-in mechanism for secure trade execution or payment settlement.

b. Weather and Advisory Applications

Platforms such as Agmarknet and other mandi price portals provide periodic updates on crop prices across regions. While these systems improve awareness of prevailing market rates, they function primarily as informational portals. They do not enable direct transactions between farmers and buyers. Moreover, price updates may not always be real-time, and there is no built-in mechanism for secure trade execution or payment settlement.

c. Agricultural Product Platforms

Some digital platforms allow farmers to list agricultural products online. However, many of these platforms are region-specific and lack scalability. In addition, they often do not provide secure authentication systems, structured order management, or real-time bidding features. Transparency and secure financial handling are frequently limited.

3. Mobile and Web-Based Farm Support Systems

Various mobile and web-based applications support farmers through educational content, extension services, and price alerts. Extension service applications provide crop management guidance but do not handle trading activities. Price alert applications notify farmers of rate fluctuations without enabling direct buyer interaction. Farmer community platforms facilitate peer communication but lack structured order processing and secure transaction systems.

These systems operate independently and are not integrated into a unified marketplace framework.

4. Gaps Identified in Existing Literature

The review of existing systems highlights several key gaps. There is no centralized digital platform that enables direct farmer-to-buyer trade with integrated authentication, real-time bidding, and secure financial settlement. Many applications lack secure online transaction handling mechanisms and do not combine real-time price reporting with marketplace functionality. Furthermore, most systems fail to integrate weather updates, demand forecasting, and trading operations within a single ecosystem.

These limitations indicate the need for a comprehensive digital platform that provides real-time pricing, secure authentication, structured order management, and transparent buyer-seller interaction while reducing dependence on intermediaries.

5. Relevance to Agri Connect

The literature survey demonstrates that current agricultural applications provide only partial solutions, such as price reporting or weather updates, but do not offer a complete transactional marketplace with real-time communication and secure settlement mechanisms. None of the existing systems integrate cloud-based infrastructure, live bidding functionality, escrow-secured payments, and scalable microservices architecture within a single platform.

AgriConnect addresses these gaps by developing a cloud-native digital marketplace that combines real-time bidding, secure authentication through Supabase, PostgreSQL-based structured databases, and WebSocket-enabled synchronization. By integrating these components, the platform creates a unified and transparent agricultural trading ecosystem aligned with modern digital transformation principles.

CHAPTER 3: SYSTEM ANALYSIS

Analysis of Existing System

• Farmers depend on local markets (mandis) and intermediaries to sell their crops.
• There is no direct and transparent communication between farmers and buyers.
• Prices are often influenced or decided by middlemen.
• No centralized cloud-based digital platform exists for structured crop trading.
• Limited access to real-time market demand and digital price discovery mechanisms.
• Manual transaction handling leads to inefficiency and delays.

Problems Identified:

- Low profit margins for farmers
- Lack of transparency in pricing
- Time-consuming manual processes
- No proper record maintenance
- Information gap between stakeholders

Need for Proposed System

• Connects farmers directly with buyers through a digital marketplace
• Enables real-time price discovery through bidding and fixed pricing
• Provides secure cloud-based authentication and data management
• Maintains structured digital transaction records
• Supports real-time communication between users
• Reduces dependency on intermediaries

AgriConnect fulfills these requirements through a cloud-native online marketplace system built on modern web technologies.

Functional Requirements

Farmer:

• Register and securely log in
• Create, update, and delete product listings
• Choose pricing method (fixed price or auction)
• View bids on listed products
• Accept winning bids (in auction model)
• Manage and track orders
• View transaction status

Buyer:

• Register and securely log in
• Search and filter products
• Participate in real-time bidding
• Place fixed-price purchase orders
• Make secure payments
• Track order and payment status

Real-Time Module:

• Synchronize bids instantly using WebSockets
• Update order and bid status dynamically
• Notify users about bid changes and order confirmations

Non-Functional Requirements

• **Performance:** Fast response time for product search and bid updates
• **Security:** Encrypted authentication and role-based access control
• **Usability:** Simple, responsive, and user-friendly interface
• **Reliability:** High system availability (designed for 99.9% uptime)
• **Scalability**: Cloud-based architecture supporting user growth
• **Data Integrity:** Secure cloud database with structured relational design

Feasibility Analysis

- **Technical Feasibility**

The system is technically feasible as it is built using modern cloud technologies such as React, Next.js, Supabase, PostgreSQL, and WebSockets. These technologies provide scalability, real-time communication, and secure authentication without requiring physical server infrastructure.

- **Economic Feasibility**

The platform uses managed cloud services, reducing infrastructure and maintenance costs. Since it operates on scalable cloud infrastructure, operational expenses are optimized and cost-effective for deployment.

- **Operational Feasibility**

The system is user-friendly and designed with intuitive dashboards for farmers and buyers. With basic digital literacy, users can easily interact with the platform. Cloud hosting ensures continuous availability and smooth operation.

System Requirements

Hardware Requirements:

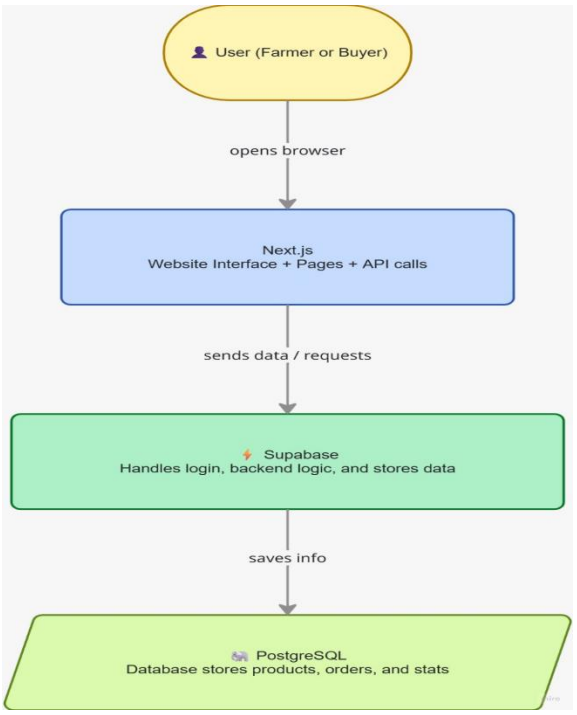
Minimum 4GB RAM
• Processor: Intel i3 or equivalent
• Stable internet connection

Software Requirements:

• Operating System: Windows, Linux, or macOS
• Web Browser: Chrome, Firefox, Edge
• Frontend: React.js with Next.js
• Backend: Supabase (Backend-as-a-Service)
• Database: PostgreSQL (Cloud-based)
• Real-Time Communication: WebSockets

CHAPTER 4: SYSTEM DESIGN

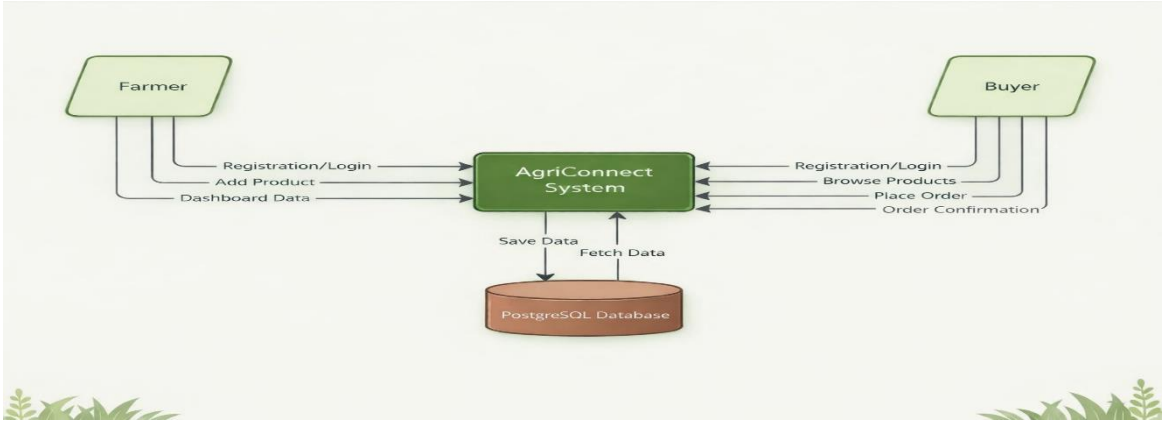
4.1 Overall System Architecture



AgriConnect follows a cloud-based three-layer architecture consisting of the user interface layer, application layer, and database layer. Farmers and buyers access the system through a web browser, where the frontend is built using Next.js and React to provide dynamic pages and API communication. The frontend sends user requests such as login, product listing, and bidding actions to the backend services handled by Supabase. Supabase manages authentication, business logic, and secure API processing. All application data, including user details, products, bids, and orders, is stored in a PostgreSQL cloud database. The architecture ensures scalability, real-time data synchronization, secure transactions, and high system availability without requiring physical server infrastructure.

4.2 Data Flow Diagram(DFD):

4.2.1 Level-1 DFD:



The Level 0 Data Flow Diagram (Context Diagram) of the AgriConnect System represents the overall interaction between external entities and the system as a single unified process. In this diagram, the

primary external entities are the Farmer and the Buyer, who interact directly with the AgriConnect System.

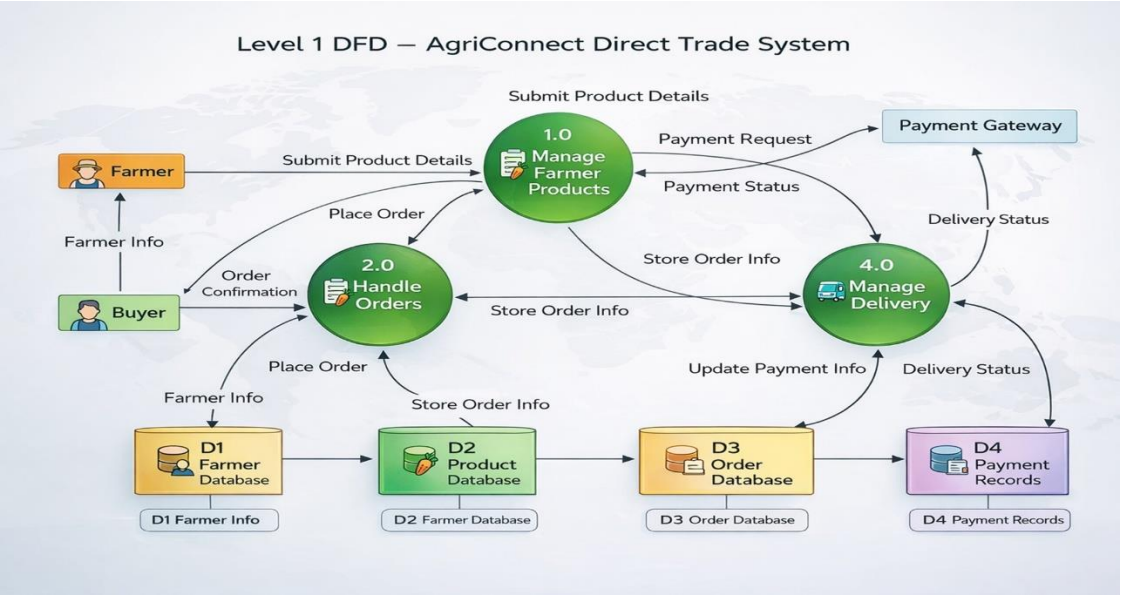
The farmer sends data to the system in the form of registration and login details, product information for adding crops, and dashboard-related requests. The system processes this information and stores it in the database. It also retrieves relevant dashboard data from the database and displays it back to the farmer.

Similarly, the buyer interacts with the system by submitting registration and login details, browsing products, and placing orders. The AgriConnect System processes these requests, fetches product information from the database, and sends order confirmation details back to the buyer.

The central process, AgriConnect System, communicates with the database to store and retrieve information. Whenever a farmer adds a product or a buyer places an order, the system saves the data in the database. When users request information such as product listings or dashboard updates, the system fetches data from the database and displays it accordingly.

Thus, the Level 0 DFD provides a high-level overview of how farmers and buyers interact with the AgriConnect platform and how the system manages data storage and retrieval to facilitate smooth agricultural trade.

4.2.2 Level-1 DFD:



The Level 1 Data Flow Diagram of the AgriConnect Direct Trade System illustrates the detailed internal processes and data interactions between users, system modules, databases, and external services. The main external entities involved in the system are the Farmer, Buyer, and Payment Gateway.

The process begins when the farmer submits product details through the “Manage Farmer Products” module. These details include crop name, quantity, pricing type, and other relevant information. The system stores this data in the Product Database (D2) and links it with the Farmer Database (D1), where farmer profile information is maintained.

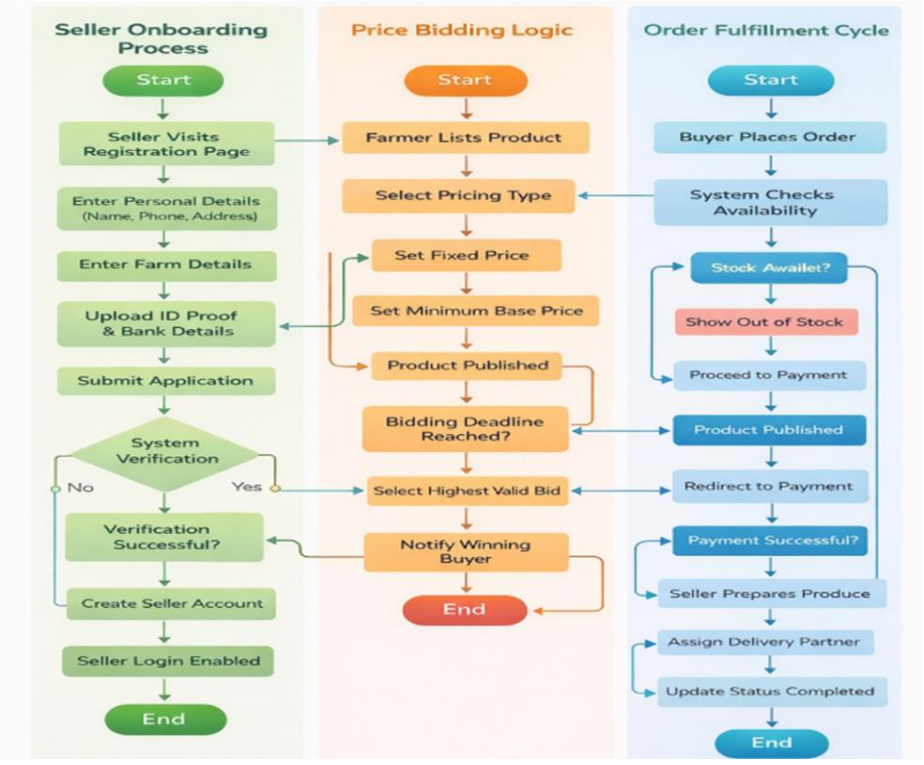
When a buyer browses the platform and places an order, the request is processed through the “Handle Orders” module. The system verifies product availability and stores the order details in the Order Database (D3). Once the order is confirmed, the buyer receives an order confirmation notification.

After order confirmation, the system generates a payment request that is sent to the Payment Gateway. The payment status returned from the gateway is recorded in the Payment Records Database (D4). This ensures secure and structured transaction management.

The “Manage Delivery” module handles shipment and delivery tracking. Delivery status updates are stored in the Order Database and linked to the corresponding payment record. Once delivery is completed, the final status is updated in the system.

Throughout the process, data flows between modules and databases in a structured manner. The Farmer Database stores user information, the Product Database stores listed crops, the Order Database maintains transaction records, and the Payment Records Database tracks payment details. This structured data flow ensures transparency, secure processing, and efficient management of agricultural trade within the AgriConnect system.

4.3 FlowChart:



The flowchart represents three main processes of the AgriConnect system: Seller Onboarding, Price Bidding Logic, and Order Fulfillment Cycle.

In the Seller Onboarding process, the farmer registers by entering personal and farm details, uploading identification and bank information, and submitting the application. The system verifies the details, and upon successful verification, a seller account is created and login access is enabled.

In the Price Bidding Logic, the farmer lists a product and selects the pricing type, either fixed price or auction. If auction mode is selected, a minimum base price is set. Once the bidding deadline is reached, the system selects the highest valid bid and notifies the winning buyer.

In the Order Fulfillment Cycle, the buyer places an order and the system checks product availability. If stock is available, the buyer proceeds to payment. After successful payment, the seller prepares the produce, assigns a delivery partner, and updates the order status to completed.

This flow ensures a structured, secure, and transparent trading process within the AgriConnect platform.

CHAPTER 5: IMPLEMENTATION

Implementation is the phase where the system design is converted into a working application. In this phase, the modules are developed, integrated, tested, and deployed.

Development Environment

The AgriConnect system is implemented using modern cloud-native web technologies. The development process follows a modular architecture where frontend, backend, and database services are logically separated but interconnected through APIs and real-time communication channels.

Frontend Implementation

• The frontend is developed using React.js with the Next.js framework.

• Tailwind CSS is used for responsive and modern user interface design.

• Dynamic routing and server-side rendering features of Next.js improve performance and SEO optimization.

• Real-time updates for bidding are handled through WebSocket connections.

• The frontend provides separate dashboards for Farmers and Buyers with interactive components and live updates.

Backend Implementation

• The backend services are managed using Supabase (Backend-as-a-Service).
• Supabase Authentication handles user registration, login, and secure session management.
• API endpoints automatically generated by Supabase manage product listings, orders, and transactions.
• Business logic for bidding, order processing, and transaction validation is implemented using server-side functions.
• Real-time synchronization of bids is achieved using WebSocket channels.

Database Implementation

• **PostgreSQL (cloud database via Supabase)** is used for structured data storage.

• The main tables include Farmers, Buyers, Products, Orders, Bids, and Payments.

• Foreign key relationships maintain data integrity between tables.

• Structured queries and real-time database listeners handle data insertion, updates, and retrieval.

• Cloud storage ensures data reliability and scalability.

Module Implementation

Farmer Module

• Secure registration and authentication using Supabase Auth.
• Product listing interface allows adding, editing, and deleting products.
• Farmers can choose between fixed pricing and auction-based pricing.
• Real-time dashboard displays active bids and order status.

Buyer Module

• Secure registration and login.
• Search and filtering functionality for products.
• Real-time bidding participation using WebSockets.
• Order placement and transaction tracking.
• Payment status updates displayed dynamically.

Real-Time Bidding Module

• WebSocket channels synchronize bid updates instantly across users.
• Highest bid tracking is automatically updated.
• Buyers receive immediate notification when outbid.
• Auction closure logic selects the highest valid bid at deadline.

Security Implementation

• Secure authentication using JWT-based Supabase Auth.
• Role-based access control for Farmers and Buyers.
• Encrypted HTTPS communication between client and server.
• Database-level access control policies to protect sensitive data.
• Secure session handling to prevent unauthorized access.

Testing and Deployment

• Each module was tested individually to verify functionality (Unit Testing).
• Integrated testing ensured seamless interaction between frontend, backend, and database.
• Real-time bidding functionality was tested under concurrent user scenarios.
• The system is deployed on cloud infrastructure, ensuring scalability and high availability without physical server maintenance.

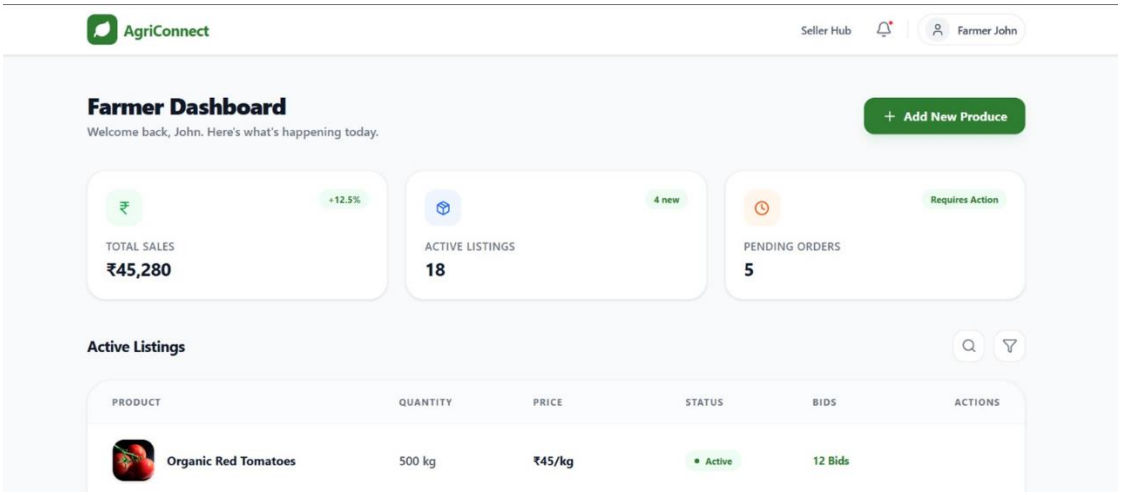
Implementation Challenges

• Designing a secure real-time bidding mechanism.
• Ensuring accurate synchronization of bid updates.
• Managing relational database integrity in cloud environment.
• Implementing secure authentication and access control.
• Optimizing frontend performance for real-time interactions.

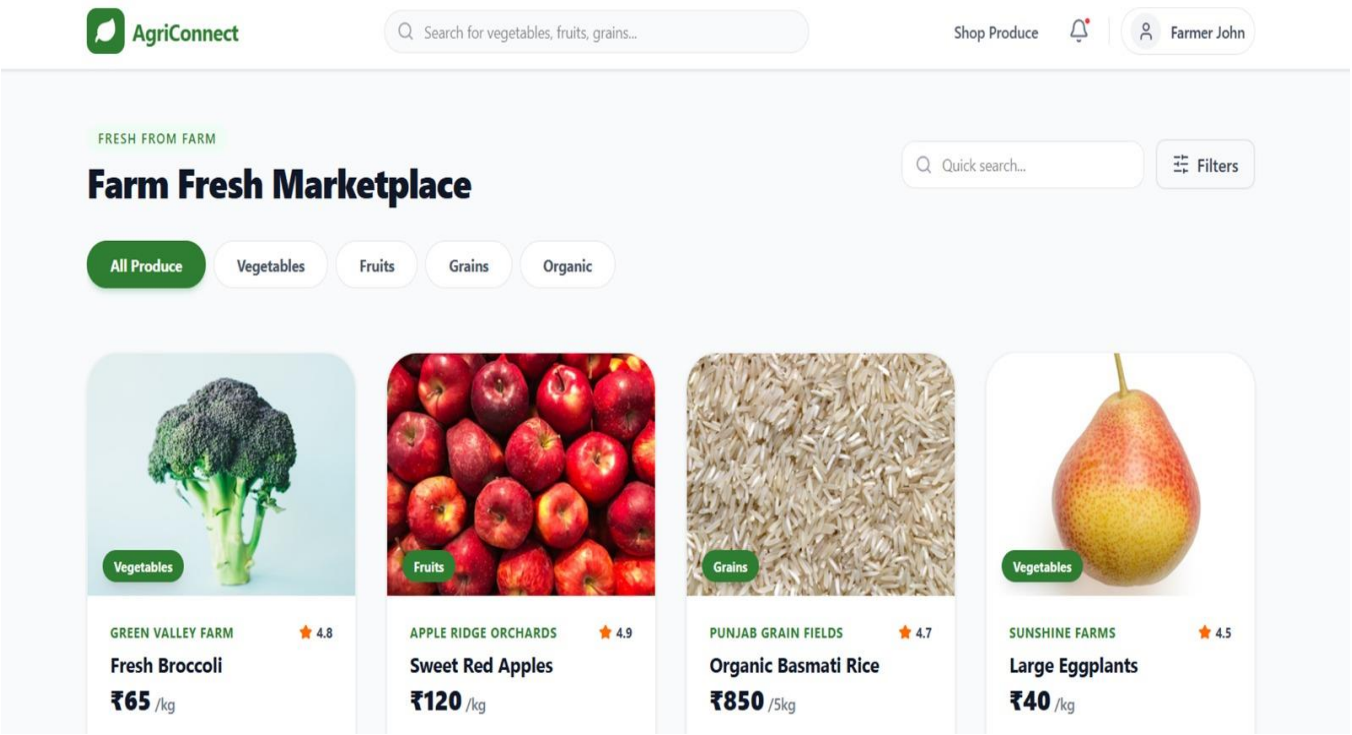
CHAPTER 6: UI/UX DESIGNS

The system successfully provides:

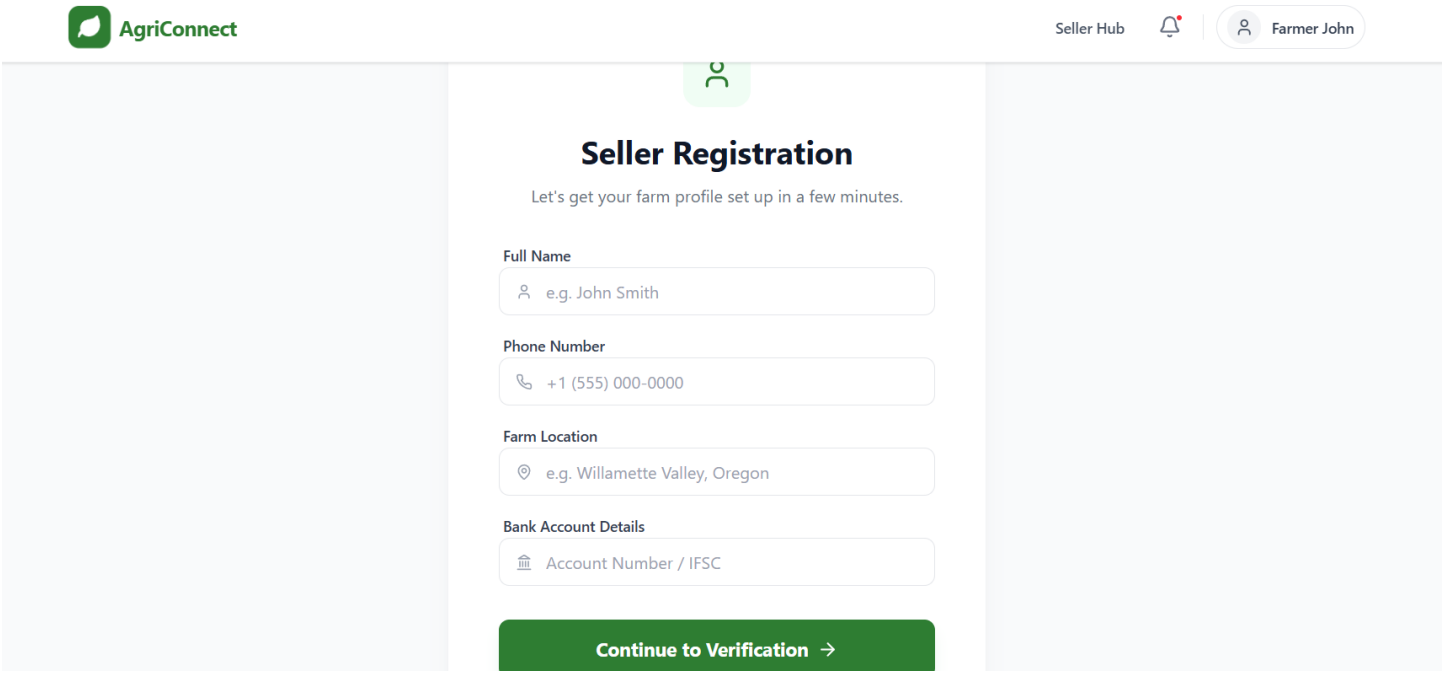
6.1 Farmer Dashboard (Add Listing)



6.2 Buyer Storefront (Search & Filter)



6.3 Seller Registration



CHAPTER 7: CONCLUSION

AgriConnect successfully addresses the key challenges present in traditional agricultural marketing systems. By providing a cloud-based digital marketplace that directly connects farmers with buyers, the platform reduces dependency on intermediaries and introduces transparent price discovery mechanisms through real-time bidding and fixed-price models.

The system enhances market accessibility, improves communication between stakeholders, and ensures secure authentication and structured data management through Supabase and PostgreSQL. The integration of real-time synchronization using WebSockets strengthens transparency and operational efficiency within the trading process.

Through this project, digital transformation is introduced into the agricultural sector, promoting income stability for farmers and reducing inefficiencies in the supply chain. AgriConnect supports sustainable agricultural development by enabling responsible trade practices and improving economic opportunities for rural communities.

In conclusion, the system demonstrates a scalable, secure, and future-ready solution for modernizing agricultural trade through cloud-native architecture and real-time digital infrastructure.

CHAPTER 8: FUTURE ENHANCEMENTS

Although AgriConnect provides a robust digital marketplace, several enhancements can further strengthen the platform.

A fully integrated escrow-based online payment gateway can be implemented to support secure digital transactions using UPI, credit/debit cards, and net banking. This will ensure automated and protected financial settlements between farmers and buyers.

A dedicated mobile application for Android and iOS platforms can improve accessibility and increase adoption, especially among rural users who primarily rely on smartphones.

Artificial Intelligence and Machine Learning models can be integrated to provide predictive crop price analysis based on historical trends, demand patterns, and seasonal variations. This would enable farmers to make more informed selling decisions.

An advanced analytics dashboard can be developed to provide insights into sales trends, demand forecasting, and revenue tracking. Such data-driven tools would enhance strategic decision-making for both farmers and buyers.

Integration with logistics and delivery tracking systems can improve order fulfillment transparency and ensure timely shipment updates. Multi-language support can be introduced to increase usability across different regions.

In future iterations, blockchain-based settlement mechanisms and government scheme integration may further improve transparency, traceability, and financial inclusion within the agricultural ecosystem.